Australian Journal of Earth Sciences (2001) **48**, 101–112

Determination of magnetic anisotropy and a *ca* 1.2 Ga palaeomagnetic pole from the Bremer Bay area, Albany Mobile Belt, Western Australia

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Granulite facies tonalitic gneiss, mafic granulite and late metadolerite dykes from Bremer Bay in the Mesoproterozoic Albany Mobile Belt yield palaeomagnetic remanence that were acquired between *ca* 1.2 Ga and 1.1 Ga. A well-constrained pole (66.6°N, 303.7°E) fits the *ca* 1.2 Ga part of the Precambrian Australian apparent polar wander path. This pole is in agreement with the high-latitude position of Australia at *ca* 1.2–1.1 Ga shown on some Rodinia reconstructions. More data are required before any significance can be attributed to a second, poorly defined pole (41.8°S, 243.7°E) that falls at some distance from the *ca* 0.8 Ga part of the Australian apparent polar wander path. Magnetic anisotropy measurements from all samples except late granite dykes indicate northeast–southwest elongation (i.e. parallel to the local trend of the orogenic belt) and northwest–southeast contraction. This is in agreement with the orientation of principal strain axes deduced from structures formed during late stages of ductile deformation. The mean magnetic fabric lineation (long axis of the strain ellipsoid) is subparallel to a mineral elongation lineation and the axes of late upright to inclined folds. Short axes of the strain ellipsoid determined from magnetic fabric measurements are in a similar orientation to poles to the axial surfaces of these folds and to the associated cleavage. This mean shortening axis bisects late conjugate ductile shear zones that overprint the folds. This study has shown that structurally complex high-grade gneisses and intrusive rocks with variable timing relationships may yield meaningful palaeomagnetic results for late stages of metamorphism. Magnetic anisotropy analysis is also seen to be a valuable tool in providing principal strain directions for late ductile deformation.

KEY WORDS: Albany Mobile Belt, gneiss, magnetic anisotropy, Mesoproterozoic, palaeomagnetism.

INTRODUCTION

Palaeomagnetic studies are generally carried out on intrusive igneous rocks (such as mafic dykes) and low-grade sedimentary rocks. High-grade metamorphic rocks, however, may also be carriers of palaeomagnetic information that corresponds to the time of exhumation and cooling (Dunlop & Ozdemir 1997). This study was aimed at determining whether meaningful palaeomagnetic results could be obtained from high-grade rocks from the Mesoproterozoic Albany Mobile Belt in southern Western Australia (Figure 1). The Bremer Bay region of the Albany Mobile Belt was chosen as its deformation history is well constrained by U/Pb SHRIMP and Rb/Sr geochronology (Black *et al.* 1992). The consistency of palaeomagnetic results of Early Cambrian dolerite dykes throughout the Albany Mobile Belt (Harris & Li 1995) has shown that there has been negligible tilting during breakup between Western Australia and Antarctica.

This study sought to determine a palaeomagnetic pole for these high-grade rocks and late dykes for comparison to the Australian apparent polar wander path. The Australian apparent polar wander path for the Mesoproterozoic (Idnurm & Giddings 1988) is presently constrained by a small number of studies, the majority of which have concerned northern and central Australia. Magnetic anisotropy was also measured in order to determine how magnetic fabric relates to regional strain and thus to indirectly establish the timing for acquisition of the magnetic fabrics.

GEOLOGY AND SAMPLING

In the Albany Mobile Belt along the southern and southeastern margins of the Archaean Yilgarn Craton (Figure 1), Archaean and Palaeoproterozoic granitoid, gneiss, metasediment and mafic dykes have been reworked within a dextral transpressional regime at approximately 1.2 Ga (Black *et al.* 1992). A Central Domain of granulite facies gneiss has been obliquely thrust onto amphibolite facies gneiss of the Northern Domain (Beeson *et al.* 1988). There is a transitional contact with the Southern Domain, which is comprised of dominantly amphibolite facies gneiss and syn- to post-tectonic granitoids. Granulite facies assemblages are locally preserved in the Southern Domain, especially in cores of regional anticlines. In reconstructions of the Rodinia supercontinent, the Albany Mobile Belt separates the Yilgarn Craton from the East

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Antarctic Shield (Harris 1995). The belt is correlated with the basement gneiss complex in Meghalaya in north-eastern India, from where it is thought to continue across India along the Central Indian Tectonic Zone (Harris 1993).

Granulite facies tonalitic and mafic gneiss of the Central Domain are best exposed in the Bremer Bay region (Figures 1, 2) in the eastern part of the belt where it trends northeast. The presence of 3.1 Ga zircon cores with 1.2 Ga zircon overgrowths and Sm–Nd model ages of *ca* 2.1 Ga (Black *et al.* 1992) in tonalitic gneiss indicates Archaean and Palaeoproterozoic precursors. High-grade metamorphism in the Central Domain granulites and intrusion of late- to post-tectonic granitoids occurred at *ca* 1.2 Ga based on U/Pb SHRIMP geochronology. The Rb/Sr system has closed at *ca* 1.1 Ga. (Black *et al.* 1992). There is a southwards progressive overprinting of Proterozoic fabrics and an increase in metamorphic grade across the Northern Domain (Beeson *et al.* 1988). Mafic horizons that are generally concordant with the regional gneissosity in the Northern Domain represent metamorphosed equivalents of dolerite dykes of unknown age in the southern Yilgarn Craton (Beeson *et al.* 1988). Isoclinally folded mafic granulite layers in the Central Domain (Figure 3a) may represent a higher grade equivalent of these dykes. The presence of a major tectonic contact between the Central and Northern Domains does not, however, allow for a direct correlation between dykes in the Yilgarn Craton and mafic granulites and no geochronology and geochemistry are available to confirm this hypothesis.

A complex deformation history during high-grade metamorphism is recorded in the Bremer Bay area. Northwest-verging folds and thrusts, along with southeast-verging 'back-thrusts', indicate northwest–southeast contraction during convergence between East Antarctica and the Yilgarn Craton. Northeast-striking normal shear zones and widespread boudinage (especially of mafic layers) indicate subsequent subhorizontal northwest–southeast extension. Quartz, plagioclase, K-feldspar, orthopyroxene and clinopyroxene ± magnetite ± biotite pegmatites, which intrude boudin necks and normal shear zones (Figure 3c)

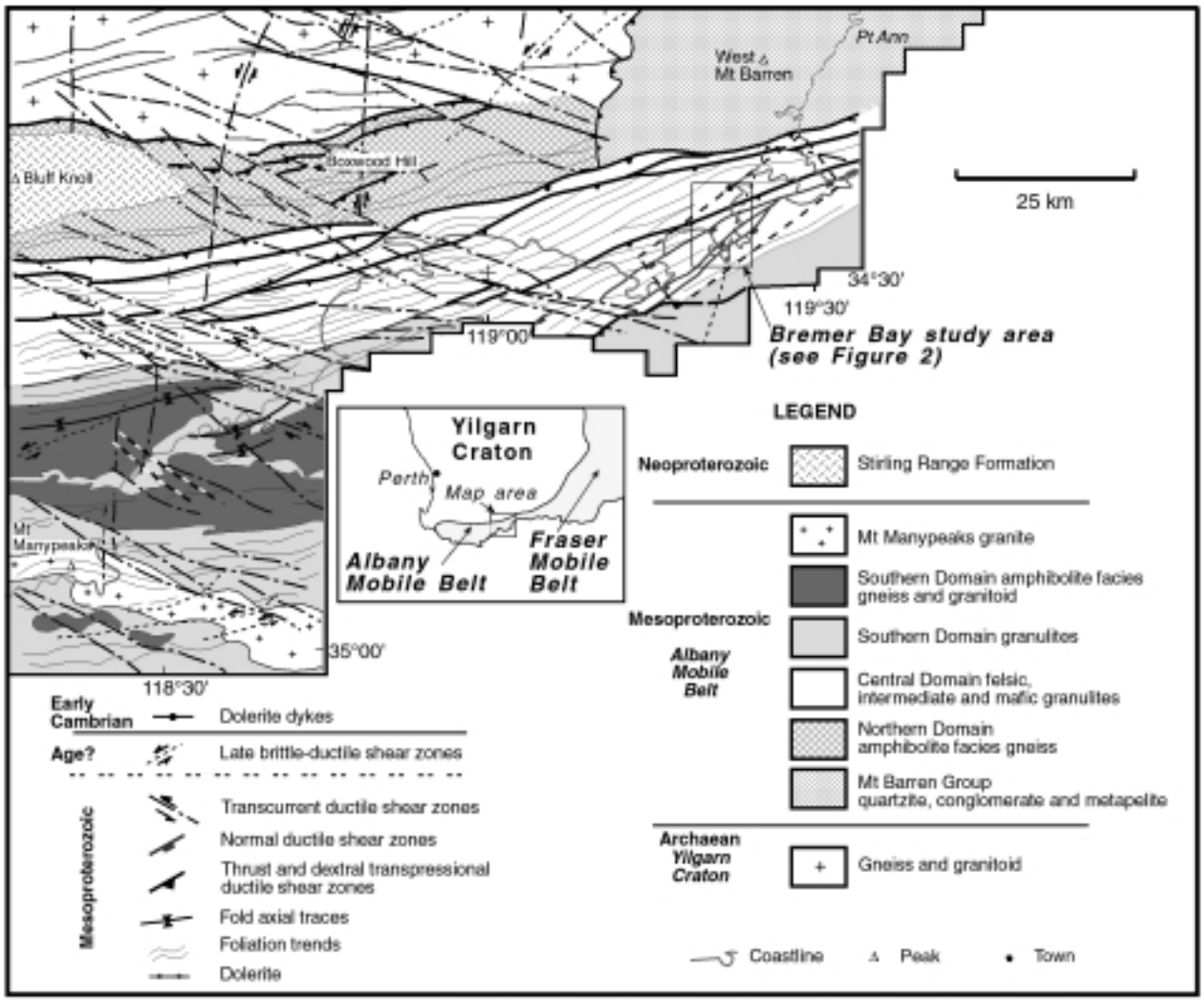


Figure 1 Structural map of the eastern Albany Mobile Belt based on interpretation of Bureau of Mineral Resources (BMR, now Australian Geological Survey Organisation) aeromagnetic data and field mapping. Inset shows regional setting. Note that the aeromagnetic interpretation extends beyond the coastline. The BMR flight line data (1.5 km line spacing, 60 m sample interval, 150 m ground clearance) were gridded at a 240 m grid cell size and artificial sun angle and vertical derivative images produced by Michael (1991a,b). The Bremer Bay study area (see Figure 2a) provides the best exposures of granulite facies gneiss of the Central Domain of Beeson *et al.* (1988).

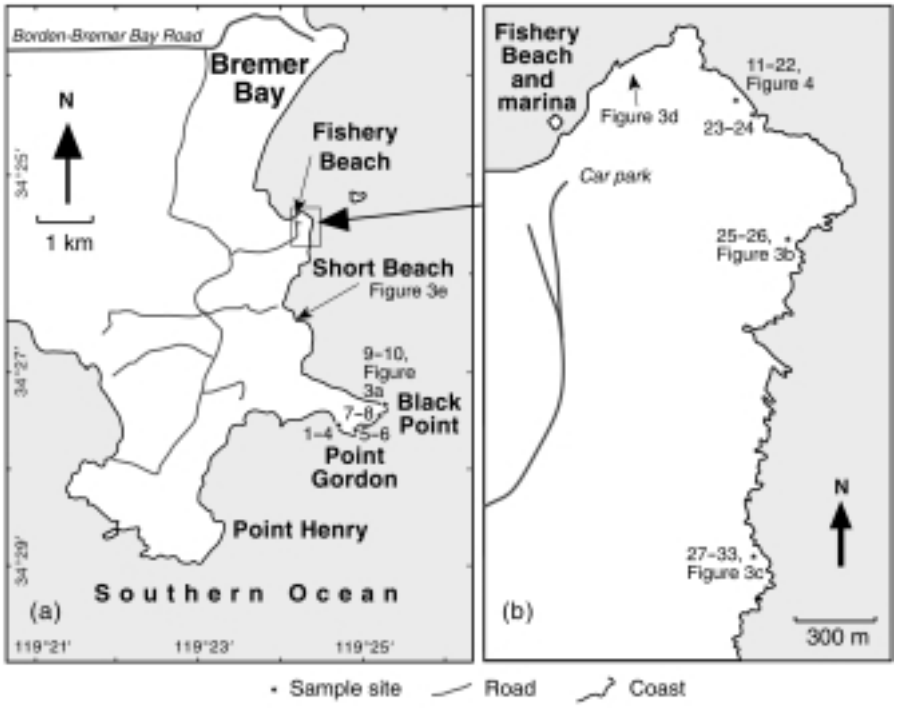


Figure 2 Maps showing location of samples and sites referred to in the text. (a) Bremer Bay region showing sample numbers for each site and location of figures. Box shows area enlarged in Figure 2b. (b) Fisheries Beach area.

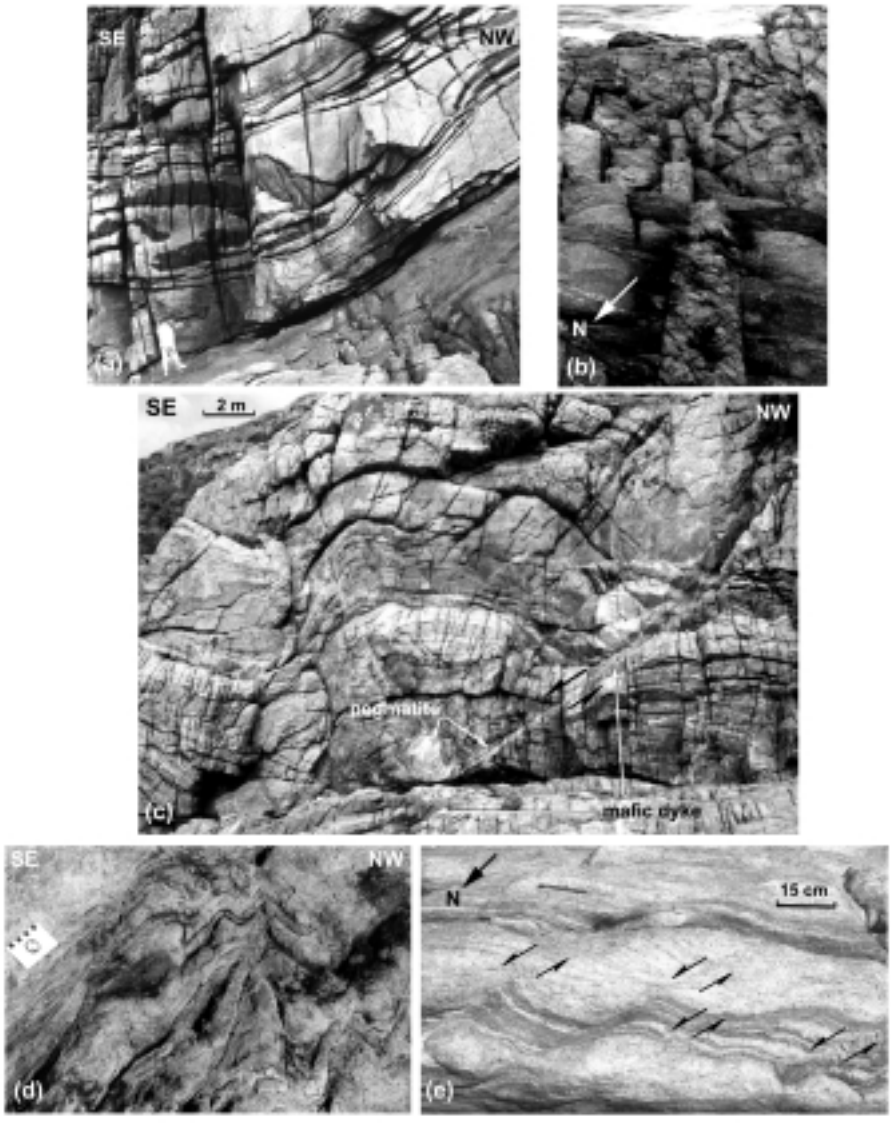


Figure 3 Structural relationships. (a) Recumbent, isoclinally folded mafic granulite layers refolded by open, upright folds above an extensional detachment. Black Point (Figure 2a). (b) Composite granitic and quartz–K-feldspar–biotite pegmatite dykes intruding late 135°-striking extension fractures. Eastern side of headland, Fisheries Beach area (Figure 2b). (c) Listric normal shear zone (intruded by pegmatite and mafic dykes) displacing refolded isoclinal folds in tonalitic gneiss and intermediate to mafic granulite. The upright antiform was initiated by rollover on the normal shear zone and tightened during subsequent northwest–southeast contraction. Eastern side of headland, Fisheries Beach area (Figure 2b). (d) Folds of gneissic foliation. Folds plunge shallowly southwest and contain an axial-planar cleavage. Folds of this generation (see Figure 12c for stereographic analysis) overprint structures formed during northwest–southeast extension and are cut by conjugate ductile transcurrent to transtensional shear zones. Scale bar in centimetre divisions. Fisheries Beach (Figure 2b). (e) Shear zones with oblique-slip (sinistral and normal) displacement. Short Beach (Figure 2a).

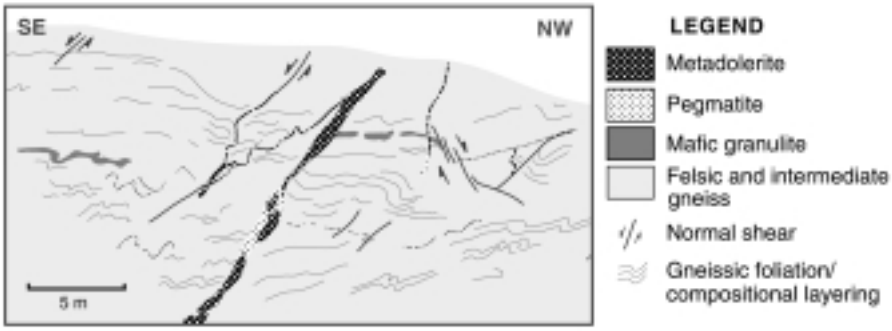


Figure 4 Northeast-striking metadolerite dykes and normal shear zones formed during northwest–southeast extension. Fisheries Beach headland (Figure 2b).

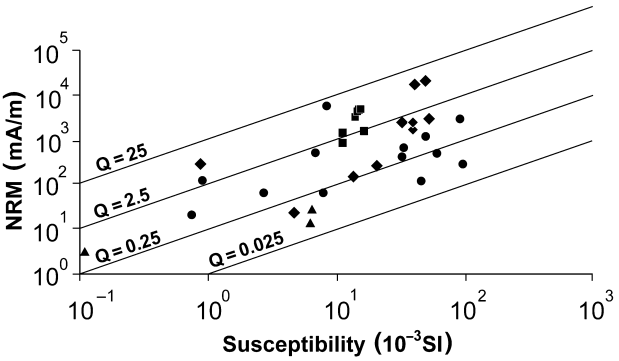


Figure 5 Remanence-susceptibility relationships for different rock types. ◆, tonalitic gneiss; ●, mafic granulite; ■, dolerite; ▲, granite; NRM, natural remanent magnetisation.

gave a Rb/Sr age of 1140 ± 40 Ma (Black *et al.* 1992). Metadolerite dykes that have intruded northeast-striking normal shear zones (Figure 3c), as discrete cross-cutting bodies (Figure 4), have been sampled for palaeomagnetic study for comparison with older mafic granulite and host gneiss. A subvertical to steeply dipping, northeast-striking spaced foliation within pegmatites and mafic dykes indicates a return to northwest–southeast contraction after their emplacement. Upright to inclined folds (Figure 3d) with an axial planar cleavage (parallel to that developed in dykes) and reverse shear zones were formed in gneisses during this later period of northwest–southeast shortening. These folds refold the early formed isoclinal folds and tighten open folds with a similar orientation developed during northwest–southeast extension due to rollover on listric normal shear zones or boudinage. Mafic boudins with northwest-trending necklines and a shallowly southwest-plunging mineral elongation lineation (subparallel to the axes of these late folds) indicate a shallow, northwest–southeast orientation of the maximum extension direction. A change in the orientation of maximum shortening from subvertical to subhorizontal during northeast–southwest extension is implied from the youngest ductile structures. Northwest-striking normal shear zones indicate subvertical shortening. Conjugate north-northeast- to northeast-striking sinistral and west-northwest- to west-striking dextral shear zones (both with a minor normal dip-slip component) are more extensively developed and indicate a maximum shortening direction plunging shallowly northwest.

The last samples come from unfoliated granitic and pegmatite dykes striking 135°. These dykes intrude

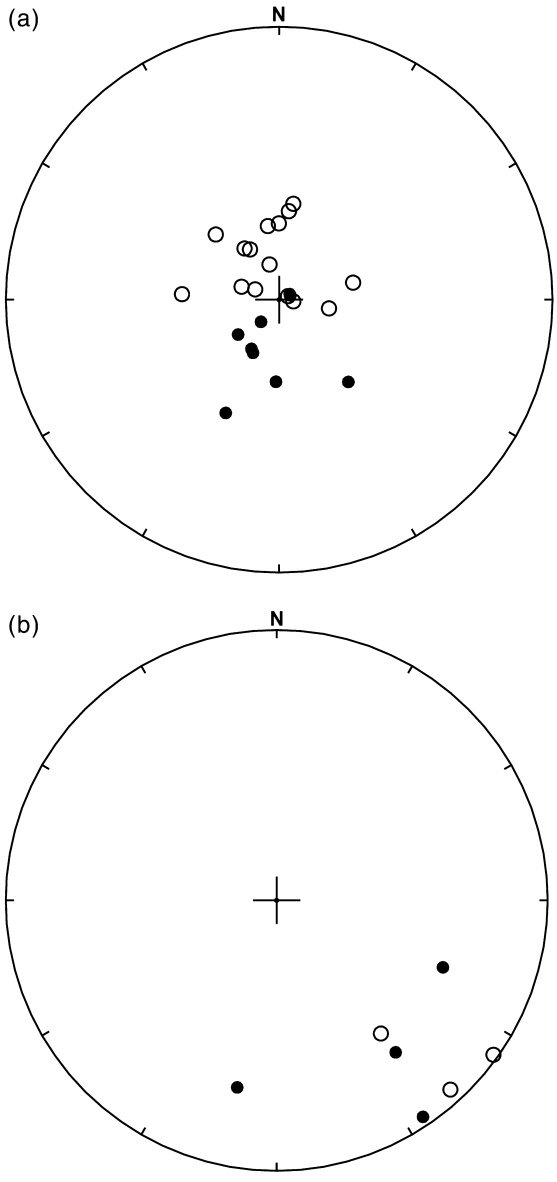


Figure 6 Equal-angle stereographic projections of sample mean palaeomagnetic directions for (a) BB1 component and (b) BB2 component. Solid and open symbols represent downward and upward directions, respectively.

extension fractures (Figure 3b) that bisect north-northwest-striking sinistral and west-northwest-striking dextral faults and brittle–ductile shear zones that define a

conjugate set. These structures suggest dyke intrusion occurred at a shallower crustal level during further northwest–southeast contraction and northeast–southwest extension. A Rb/Sr date of *ca* 1.1 Ga has been obtained for these dykes, although this may not represent a true intrusive or metamorphic age due to their coarse grainsize (Black *et al.* 1992).

LABORATORY METHODS AND MAGNETIC PROPERTIES

Thirty-three block-orientated samples were collected from mafic and tonalitic granulite, metadolerite and granitic dykes, using a sun compass for sample orientation. Eighty-eight cylindrical specimens (25 mm diameter) were drilled from the sampled blocks. The palaeomagnetic and magnetic anisotropy study has been carried out in the palaeomagnetic laboratory of the Tectonics Special Research Centre at the University of Western Australia. Magnetic remanence was measured using a 2G755R cryogenic magnetometer. At least one specimen from each block was subjected to stepwise thermal demagnetisation using MMTD2 furnaces manufactured by Magnetic Measurements (~10 nT residual field). Also at least one specimen from each block was subjected to stepwise alternating field (AF) demagnetisation using a 2G600 automated degaussing system. The results obtained from the pilot collection were used for determining the demagnetisation intervals and for the number of steps to be used for the remaining specimens. Magnetic susceptibility was measured after each heating step with a MS2 susceptibility meter. Isothermal remanent magnetisation (IRM) experiments were carried out using a MMPM9 pulse magnetiser.

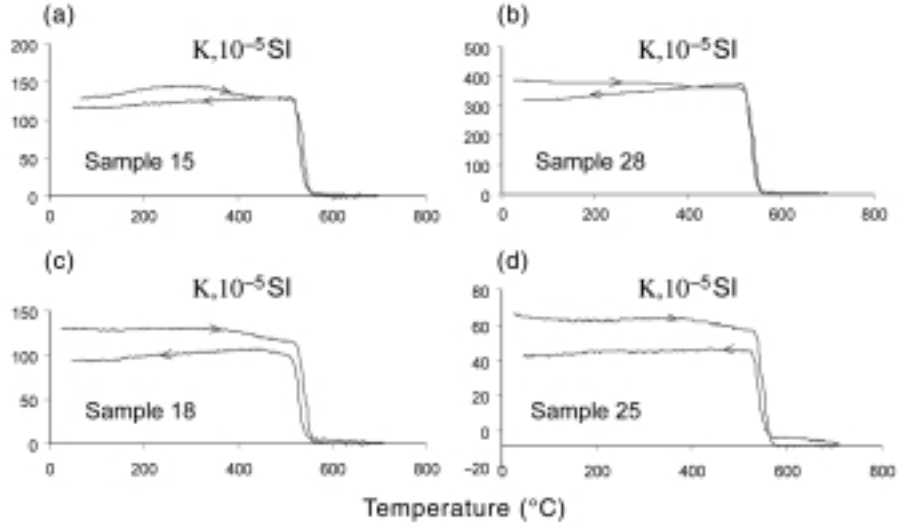


Figure 7 Examples of thermomagnetic curves. (a) Tonalitic gneiss. (b) Mafic granulite. (c) Dolerite. (d) Granite.

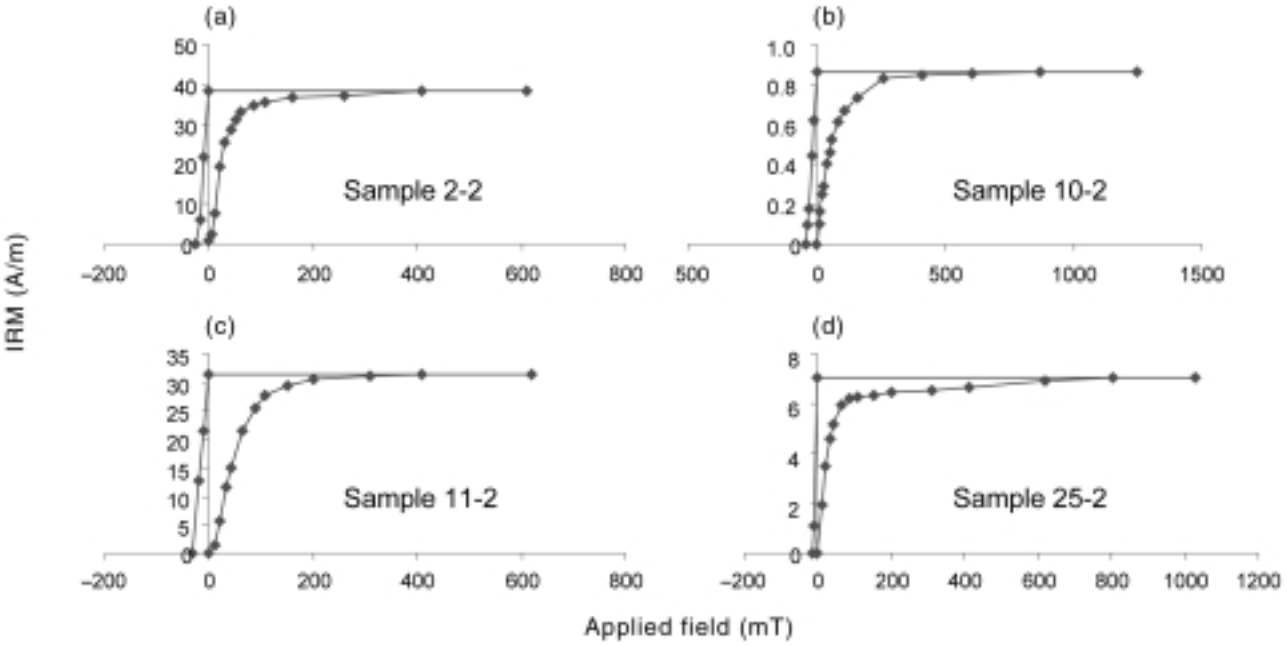


Figure 8 Examples of isothermal remanent magnetisation (IRM) saturation curves. (a) Tonalitic gneiss. (b) Mafic granulite. (c) Dolerite. (d) Granite.

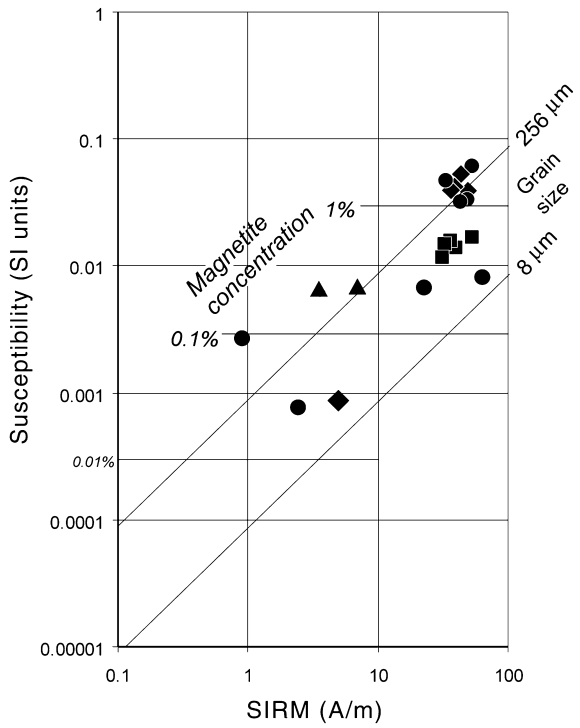


Figure 9 Bilogarithmic saturation isothermal remanent magnetisation (SIRM) *vs* susceptibility plot. The concentration/grainsize grid is for pure magnetite according to Thompson and Oldfield (1986). ◆ tonalitic gneiss; ●, mafic granulite; ■, dolerite; ▲, granite.

Initial measurements of natural remanent magnetisation (NRM) and magnetic susceptibility showed large variations (Figure 5). The ratio of remanent magnetisation to induced magnetisation (Koenigsberger's Q parameter) usually reflects the grainsize distribution of the magnetic material and, therefore, how magnetically soft particles are, or whether the remanence is a thermal remanence magnetisation (TRM) or an IRM, because high values of Q are often generated by IRM caused by lightning. Only dolerites define a compact group with relatively high values of Q between 1.94 and 8.02. Such values indicate that these rocks have stable remanences (Piper 1987 p. 17). The probability of lightning remagnetisation is not very high

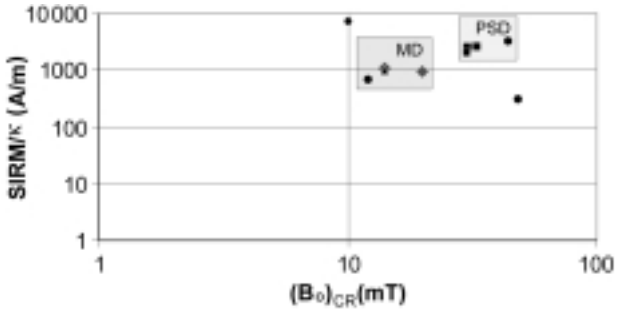


Figure 10 Bilogarithmic saturation isothermal remanent magnetisation (SIRM)/κ *vs* coercivity of remanence [$(BO)_{CR}$] plot. Multidomain (MD) and pseudo-single-domain (PSD) magnetite areas are shown according to Thompson and Oldfield (1986).

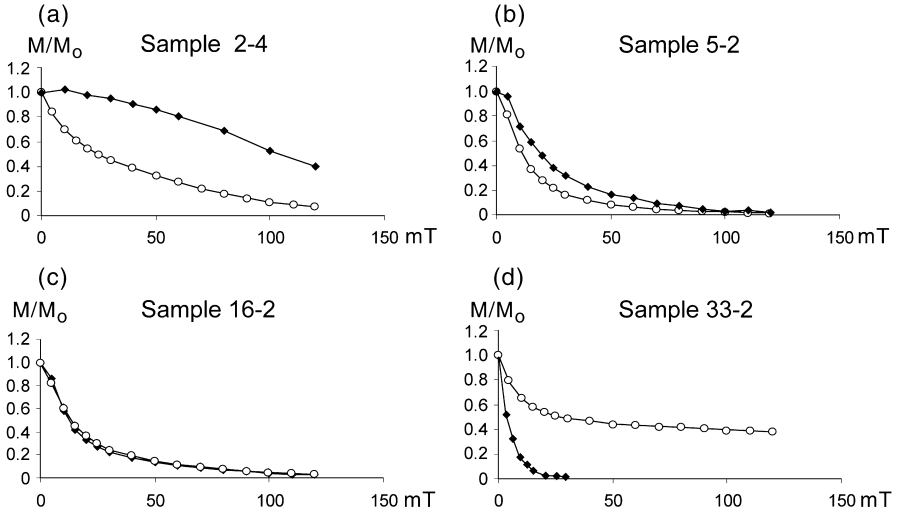


Figure 11 Examples of the comparison of alternating field (AF)-demagnetisation curves of saturation isothermal remanent magnetisation (SIRM, ○) and natural remanent magnetisation (NRM, ◆)—the version of the Lowrie–Fuller test. (a) Tonalitic gneiss. (b) Mafic granulite. (c) Dolerite. (d) Granite.

Table 1 Mean palaeomagnetic directions from metamorphic rocks from Bremer Bay, Western Australia (34.0°S, 119.5°E).

Component of magnetisation	*N*/*n*	Decl. (°)	Incl. (°)	k	α_{95} (°)	Plat (°N)	Plong (°E)	D_p (°)	D_m (°)
1 BB1, positive incl.	9/17	190.9	70.2	14.4	14.1	69.0	281.5	20.9	24.3
2 BB1, negative incl.	15/33	347.0	–73.4	13.5	10.8	63.9	314.7	17.4	19.4
3 BB1, N + R	24/50	176.9	72.5	14.0	8.2	66.6	303.7	12.9	14.5
4 BB2	7/12	141.9	3.6	6.6	25.5	–41.8	243.7	12.8	25.5

N/*n*, number of samples/specimens; Decl, Incl, sample mean declination and inclination, respectively; k, best estimate of the precision parameter of Fisher (1953); α_{95}, the semi-angle of the 95% cone of confidence; Plat, Plong, latitude and longitude of the palaeopole, respectively; D_p, D_m, the semi-axes of the cone of confidence about the pole at the 95% probability level.
BB1, Bremer Bay 1; BB2, Bremer Bay 2.

in our case due to relatively small scatter of the remanence directions (Figure 6). The same range of Q values has been found for most of the studied tonalitic gneisses, although three samples carry smaller values of Q. Magnetic properties of mafic granulites vary widely, and seven samples show Q > 0.5. Granites are relatively weakly magnetised and their Q values are low. The low-field susceptibility *vs* temperature curves for specimens from the various rock types (Figure 7) all point to a narrow Curie temperature range close to that of pure magnetite (580°C: Dunlop &

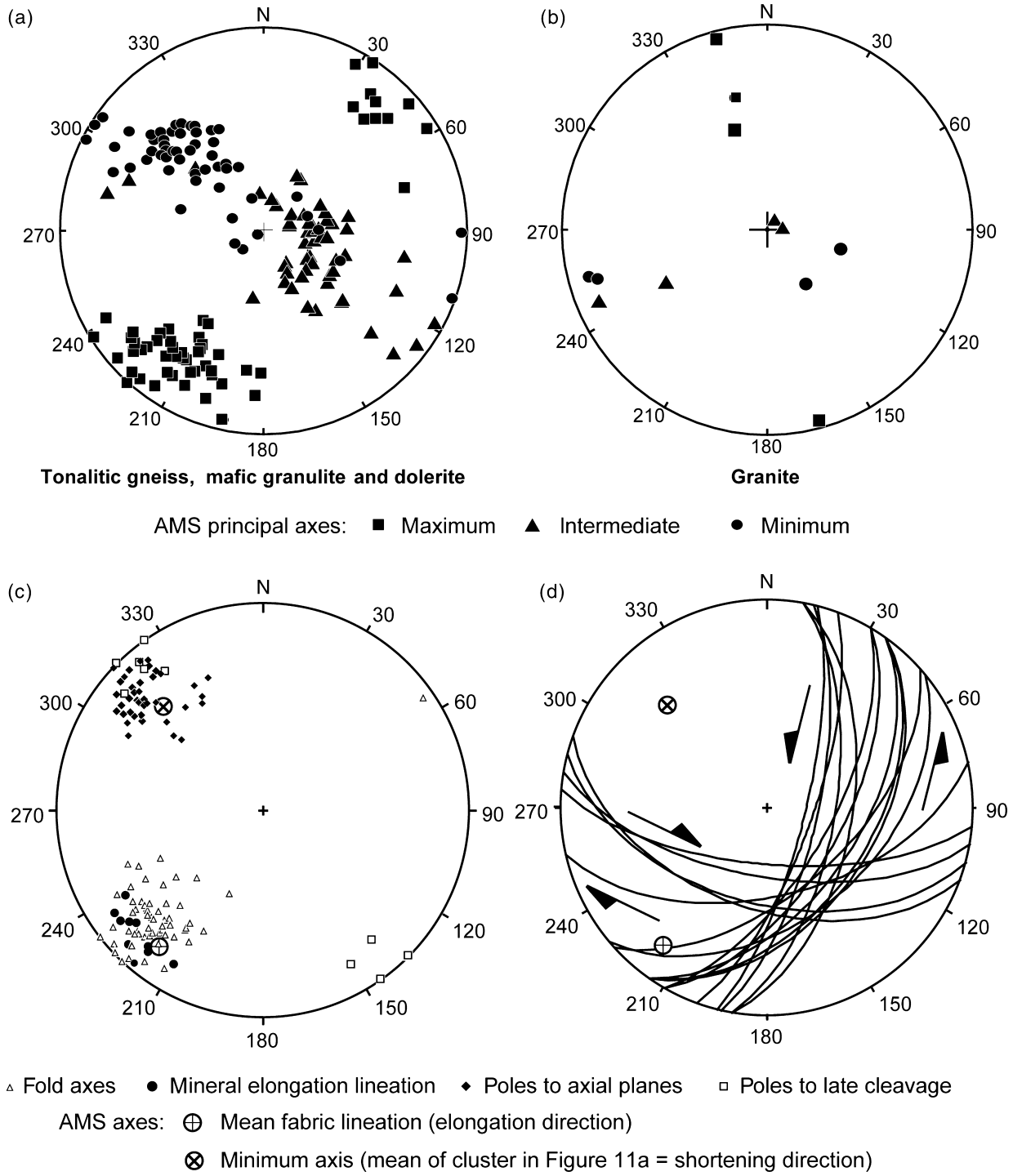
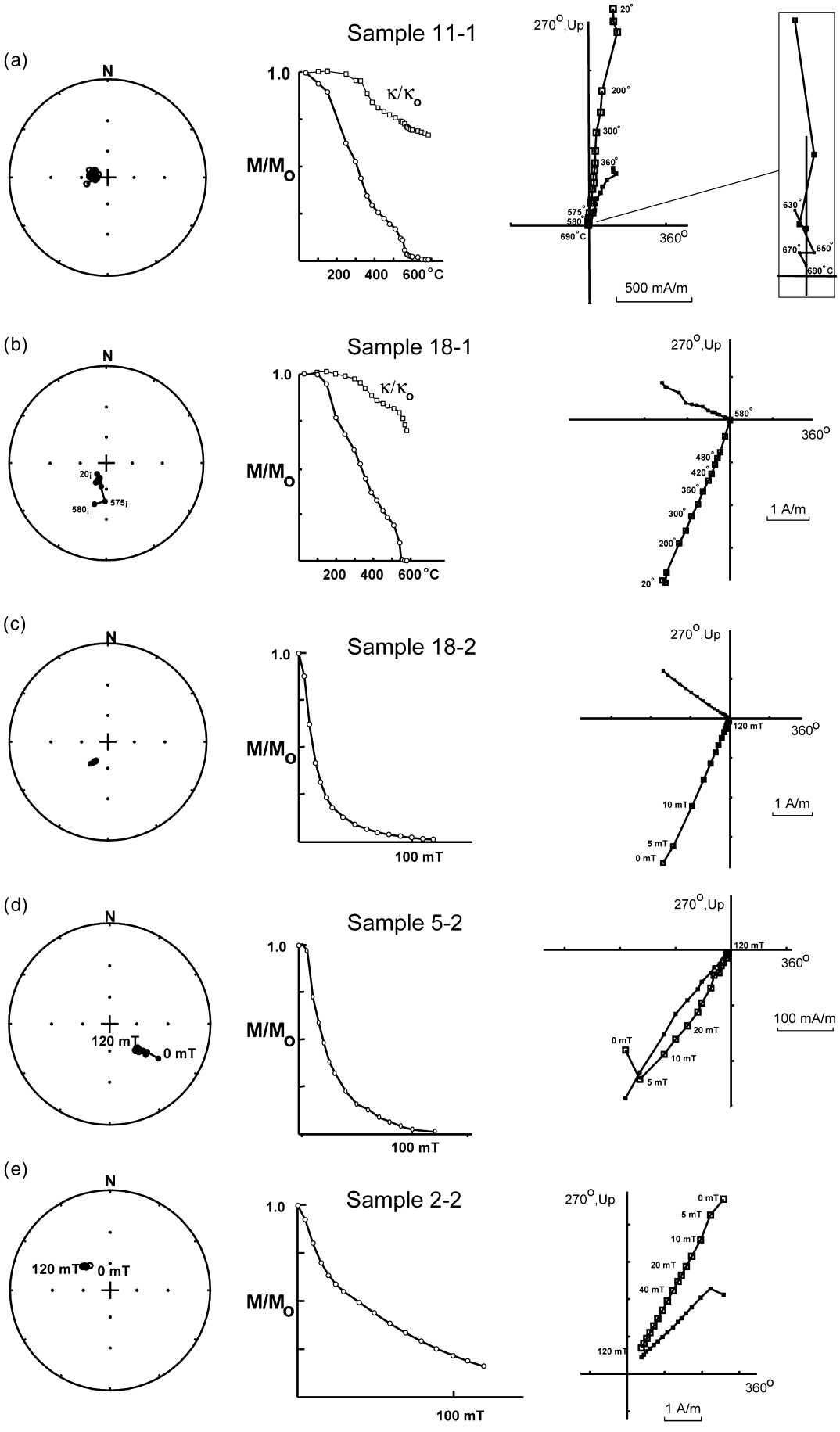


Figure 12 (a, b) Equal-angle stereographic projection of the anisotropy of magnetic susceptibility (AMS) principal axes (*in situ*). (c) Stereographic projection of mineral elongation lineations and axes and poles to axial surfaces for the youngest generation of folds. Note that the magnetic fabric lineations in (a) are subparallel to the mineral-elongation lineations and fold axes. The cluster of minimum fabric axes determined from AMS measurements is similar in orientation to the poles to the measured cleavage and axial surfaces to late folds. The mean magnetic fabric lineation and mean of the cluster of minimum principal magnetic fabric axes are indicated. Fisheries Beach area (see Figure 2 for location). (d) Stereographic projection of late ductile shear zones (great circles) that form a conjugate set. Their orientations are consistent with formation during shortening subparallel to the mean minimum AMS principal axis and elongation subparallel to the mean magnetic fabric lineation. Short Beach (see Figure 2a for location).


(a)
N
Sample 11-1
1.0
κ/κo
M/Mo
200 400 600°C
270°,Up
20°
200°
300°
360°
575°
580°
690°C
360°
630°
670°
650°
690°C
500 mA/m
(b)
N
20i
580i
575i
Sample 18-1
1.0
κ/κo
M/Mo
200 400 600°C
270°,Up
580°
360°
480°
420°
360°
300°
200°
20°
1 A/m
(c)
N
Sample 18-2
1.0
M/Mo
100 mT
270°,Up
120 mT
360°
10 mT
5 mT
0 mT
1 A/m
(d)
N
120 mT
0 mT
Sample 5-2
1.0
M/Mo
100 mT
270°,Up
120 mT
360°
0 mT
20 mT
10 mT
5 mT
100 mA/m
(e)
N
120 mT
0 mT
Sample 2-2
1.0
M/Mo
100 mT
270°,Up
0 mT
5 mT
10 mT
20 mT
40 mT
120 mT
360°
1 A/m

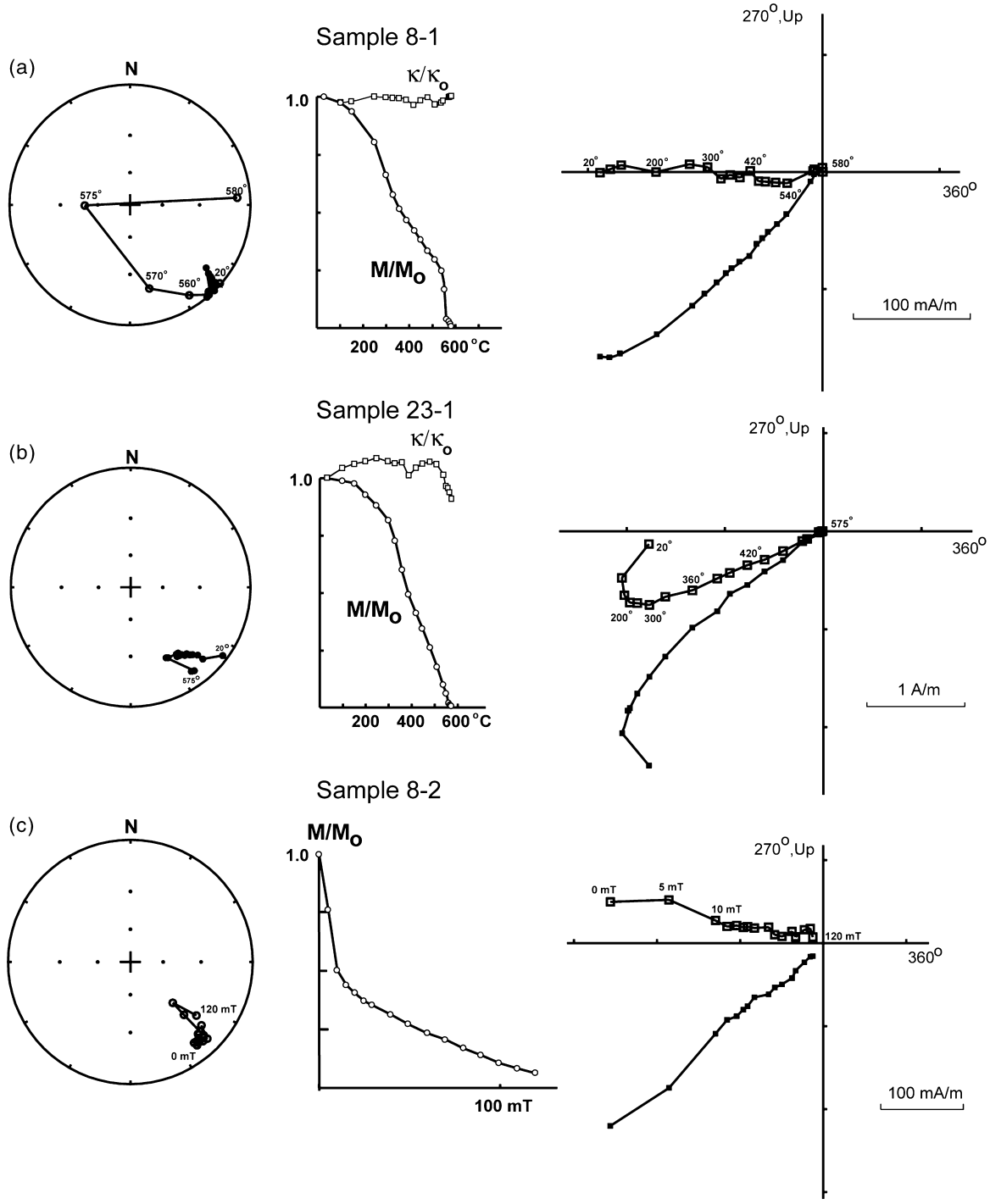


Figure 14 Examples of thermal and alternating field (AF) demagnetisation of rocks showing the BB2 component. (a–c) Mafic granulite. See Figure 13 for legend.

Figure 13 Examples of thermal and alternating field (AF) demagnetisation of rocks showing the BB1 component. Equal-angle projections are shown in the first column, normalised magnetisation and susceptibility decays in the second column, and orthogonal Zijderveld plot in the third column. (a–c) Dolerite. (d) Mafic granulite. (e) Tonalitic gneiss. Solid and open symbols on the stereographic projection represent downward and upward directions, respectively; solid and open symbols on the Zijderveld (1967) diagram denote horizontal and vertical projections, respectively.

Ozdemir 1997 table 3-1). Curves do not show any sharp increase of susceptibility in approaching the Curie temperature (Hopkinson's peaks), which indicates the dominance of pseudo-single-domain (PSD) and multi-domain (MD) magnetite (Schmidt 1993). Minor features on heating curves at approximately 350°C are probably caused by the oxidation of a small amount of maghemite. The fact that cooling curves show a small decrease of magnetic susceptibility after heating supports this suggestion.

Saturation IRM (SIRM) curves show similar coercivity properties for all studied rock types (Figure 8). Despite

the dominance of MD magnetite, IRM also increases between ~100 and 500 mT. This may be caused by the presence of single-domain (SD) and/or PSD magnetite (Lowrie 1990; Dunlop & Ozdemir 1997). The magnetic susceptibility *vs* SIRM diagram (Figure 9) proposed by Thompson and Oldfield (1986) shows that MD magnetite dominates in many cases. However, the estimated grainsize of magnetic particles in dolerites, some tonalitic gneisses and some mafic granulites, is inside or close to the PSD range. Thompson and Oldfield (1986) also proposed to use an SIRM *vs* remanence coercivity [$(BO)_{CR}$] plot to determine the structural state in magnetic mixtures. All treated dolerites and one sample of mafic granulite fall into the PSD area (Figure 10).

Figure 11 shows some examples of the comparison of AF demagnetisation curves for NRM and SIRM. This is the version (Dunlop & Ozdemir 1997) of the Lowrie–Fuller test (Lowrie & Fuller 1971). This test is often used to determine grainsizes and domain states present in rocks. Although the test is not as definitive as would be liked, particularly on synthetic materials, it is appealing because it is simple, and combined with other evidence further constrains interpretation. The Lowrie–Fuller test predicts that for fine (SD/PSD) particles, a low-field TRM (i.e. an NRM if acquired in the Earth's field during cooling) is more stable than a high-field TRM, approximated here by using an SIRM. For coarse (MD) particles this relationship is reversed. Therefore, a simple comparison of the demagnetisation of NRM followed by repeat demagnetisation of SIRM, indicates the predominant domain state of the magnetic particles in a sample. In most cases for our study it gave a zero result (Figure 11c). Figure 11a (tonalitic gneiss) and Figure 11b (mafic granulite) show two exceptions that indicate the SD behaviour. The typical MD result for granite is shown on Figure 11d.

All these analyses indicate that the studied metadolerites, mafic granulites and tonalitic gneisses are suitable for palaeomagnetic analysis as they contain a sufficient amount of magnetite not only with MD, but also with PSD and SD structure, and therefore may carry a stable ancient remanence. Further demagnetisations also indicated the presence of hematite in at least some of these rocks. It appears unlikely that the granites studied contain some stable remanence.

MAGNETIC FABRIC

Anisotropy of magnetic susceptibility (AMS) was measured in all specimens. The degree of anisotropy is high in the great majority of cases. The best parameter to characterise this is the parameter P_j of Tarling and Hrouda (1993 p. 21), which is defined as:

$$P_j = \exp\sqrt{\{2[(\eta_1-\eta_m)^2 + (\eta_2-\eta_m)^2 + (\eta_3-\eta_m)^2]\}}$$

where $\eta_1 = \ln K_1$, $\eta_2 = \ln K_2$, $\eta_3 = \ln K_3$, $\eta_m = (\eta_1 + \eta_2 + \eta_3)/3$, and K_1, K_2 and K_3 are the principal susceptibilities. In the Bremer Bay samples, P_j ranges from 1.1 to 12.0, with a median value of 1.301. Stereographic projections of the AMS principal axes are shown on Figure 12a, b.

Similar principal axes of anisotropy (Figure 12a) were found for all rock types except granite (Figure 12b). Agreement of magnetic fabric determinations for rocks

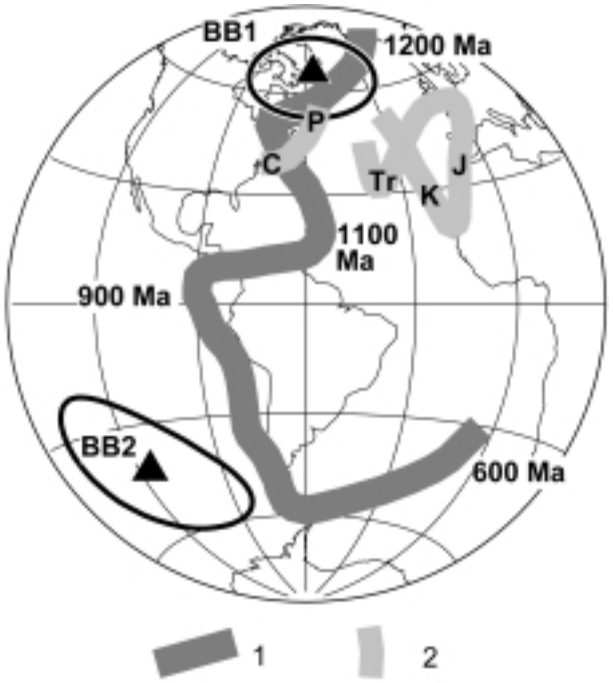


Figure 15 Palaeomagnetic poles from Bremer Bay rocks (BB1 and BB2). 1, fragment of the Precambrian apparent polar wander path for Australia (Idnurm *et al.* 1995); 2, fragments of the Phanerozoic apparent polar wander path for Australia (Harris & Li 1995). C, Carboniferous; P, Permian; Tr, Triassic; K, Cretaceous; J, Jurassic.

with different structural and intrusive relationships suggests that no pre-metamorphic magnetic fabric survived the high-grade metamorphism. Well-grouped directions of the maximum axes define the fabric lineation (Tarling & Hrouda 1993; Borradaile & Henry 1997), which has a mean direction of 19° towards 217°. The directions of intermediate and minimum axes form a girdle that is characteristic for a prolate magnetic fabric ellipsoid. However, the concentration of a large group of minimum axes about 32° towards 316° suggests the presence of a foliation perpendicular to this direction. The magnetic fabric is in close agreement with structural observations (Figure 12 c, d). Figure 12c shows a close agreement between the foliation orientation indicated from AMS measurements and the late cleavage (defined by the alignment of biotite and accentuated by subsequent jointing along it) and axial surfaces to the last generation of upright to inclined folds, such as in Figure 2d. The mean fabric lineation is also subparallel to the observed mineral elongation lineation and axes of late folds (Figure 12c) and is normal to the northwest-trending boudin axes. The orientation of late conjugate ductile shear zones (as shown in Figure 2e) is consistent with northeast–southwest extension and the dominant shallowly northwest-plunging minimum fabric axes (orientation of maximum shortening) determined from AMS measurements (Figure 12d). The northwest strike of the younger normal shear zones is again in agreement with a shallow, northeast–southwest maximum extension direction. Normal shearing is consistent with AMS measurements that show a steeper orientation of minimum fabric axes (Figure 12a).

Three specimens from granite dykes (as shown in Figure 3b) show quite different directions of AMS axes (Figure 12b). This may be due to the younger age of these dykes, which post-date penetrative deformation. However, more samples must be studied to confirm this suggestion and to determine the regional structural significance of the magnetic fabric.

PALAEOMAGNETIC ANALYSIS

Thermal and AF demagnetisations revealed two characteristic components of NRM (ChRM)–BB1 and BB2 (Table 1). Not a single sample contains both of these components. Only in granites is a stable remanence absent. A BB1 bipolar steeply inclined component has been isolated in 24 samples of tonalitic gneiss, dolerite and mafic granulite (Figure 13). Figure 13a, e shows that in those cases when the hematite component was isolated, its direction also coincides with the magnetite one. No correlation between polarity and rock type was found. Moreover, both polarities are present in the late, cross-cutting dolerite dyke (Figure 13a–c). This one-component, bipolar stable remanence (Figure 6a) in all studied rock types of the area (except granites) was probably acquired in the process of slow cooling during the uplift at the last stage of the metamorphic event. A few samples from contacts of the dolerite dyke and older mafic bodies also have the BB1 direction. This can also be explained by a metamorphic overprint. The reversal test of McFadden and McElhinny (1990) has been used. This test calculates the angle between the mean direction for normal polarity and the mean direction for reversed polarity rotated by 180°(γ). It also calculates the angle (γ_c) at which the two directions become significantly different at the 95% confidence level. The test is positive if $\gamma < \gamma_c$, but it is classified as A if $\gamma_c \leq 5°$, as B if $\gamma_c \leq 10°$, and C if $\gamma_c \leq 20°$. In our case, this gave a positive result for the BB1 component with a classification of C.

A BB2 single-polarity component has been isolated in seven samples of mafic granulites (Figure 14). No systematic differences in magnetic characteristics [NRM, magnetic susceptibility, Q ratio, SIRM and $(BO)_{CR}$] of these samples and mafic granulites with BB1 direction have been found. BB2 component is rather scattered (Figure 6b) and absent in tonalitic gneiss, dolerite and granite. Mean directions, statistics and pole positions for BB1 and BB2 components are given in Table 1.

DISCUSSION AND CONCLUSIONS

Figure 15 shows the palaeomagnetic poles for BB1 and BB2 components together with the 1200–600 Ma fragment of the Australian apparent polar wander path (Idnurm & Giddings 1988; Idnurm *et al.* 1995) and fragments of the Phanerozoic Australian apparent polar wander path (Harris & Li 1995). The BB1 pole (66.6°N, 303.7°E) fits the *ca* 1.2 Ga part of Precambrian Australian apparent polar wander path. However, this particular part is poorly determined as some of the earlier poles it is based on have been revised (Tanakah & Idnurm 1994) and other poles have poor age constraints. The closeness of this part of the Australian apparent polar wander path to the Late Palaeozoic Australian apparent polar wander path must also be considered. No Late Palaeozoic thermal or chemical event that could cause the magnetic overprint, however, is known for the Bremer Bay area. Although there has been local dextral strike-slip reactivation of probable Early Cambrian age (Harris & Li 1995) in areas of steeply dipping, north-east-striking gneissic foliations, formation of cataclastic fabrics in this event suggests deformation occurred at a shallow level in the crust. The temperature in this late brittle event would therefore be expected to have then been lower than the Curie temperature, so that resetting would not occur. However, no late penetrative cataclastic fabrics were developed in the areas sampled due to the shallower dip of the original gneissosity. Dolerite dykes that intruded in some other parts of the Albany Mobile Belt in an Early Cambrian event (Harris & Li 1995) are also absent in the Bremer Bay region. Magnetic anisotropy measurements are also in agreement with the Mesoproterozoic deformation, but are inconsistent with the deformation regime interpreted for the Early Cambrian event (east-south-east–west-northwest shortening, north-northeast–south-southwest extension: Harris & Li 1995). All these factors lead us to conclude that the palaeomagnetic pole records the *ca* 1.2–1.1 Ga metamorphic event. Given its Precambrian origin, the steep inclination of the BB1 component also supports the high-latitude position of Australia at approximately 1200–1100 Ma shown on some Rodinia reconstructions (Gose *et al.* 1997).

The BB2 pole (41.8°S, 243.7°E) falls at some distance from the *ca* 0.8 Ga part of the Australian apparent polar wander path (Figure 15). However, this palaeomagnetic direction has poor statistics (Figure 6b, Table 1) and is based on seven samples only, so an additional study is necessary to confirm (or reject) it before any tectonic interpretations can be made.

The consistency of the palaeomagnetic BB1 pole from the Albany Mobile Belt with the Australian apparent polar wander path illustrates that high-grade rocks from within an orogenic belt with a complex deformation and intrusive history may yield meaningful palaeomagnetic results. The magnetic anisotropy is also in close agreement with structural data. This study thus opens the way for more research in the Albany Mobile Belt and in similar orogenic belts (where palaeomagnetic work has not been previously attempted) on palaeomagnetic determinations and the use of magnetic fabric for structural analysis.

ACKNOWLEDGEMENTS

Fieldwork in the Albany Mobile Belt was funded by the Australian Research Council. SAP acknowledges support through a University of Western Australia Gledden Senior Visiting Fellowship. P. Schmidt and A. Whitaker are thanked for comments on an earlier draft. This is Tectonics Special Research Centre Publication Number 111.

REFERENCES

Beeson J., Delor C. P. & Harris L. B. 1988. A structural and metamorphic traverse across the Albany Mobile Belt, Western Australia. *Precambrian Research* **40/41**, 117–136.

BLACK L., HARRIS L. B. & DELOR C. P. 1992. Reworking of Archaean and Early Proterozoic components during a progressive, Middle Proterozoic tectonothermal event in the Albany Mobile Belt, Western Australia. *Precambrian Research* **59**, 95–123.

BORRADAILE G. J. & HENRY B. 1997. Tectonic applications of magnetic susceptibility and its anisotropy. *Earth Science Reviews* **42**, 49–93.

DUNLOP D. J. & OZDEMIR O. 1997. *Rock Magnetism*; *Fundamentals and Frontiers*. Cambridge Studies in Magnetism 3. Cambridge University Press, Cambridge.

GOSE W. A., HELPER M. A., CONNELLY J. N., HUTSON F. E. & DALZIEL I. W. 1997. Paleomagnetic data and U–Pb isotopic age determinations from Coats Land, Antarctica: implications for late Proterozoic plate reconstructions. *Journal of Geophysical Research* **102**, 7887–7902.

HARRIS L. B. 1993. Correlations between the Central Indian Tectonic Zone and the Albany Mobile Belt of Western Australia: evidence for a continuous Proterozoic orogenic belt. *In*: Findlay R. H., Unrug R., Banks M. R. & Veevers J. J. eds. *Gondwana 8: Assembly Evolution and Dispersal*, pp. 165–180. A. A. Balkema, Rotterdam.

HARRIS L. B. 1995. Correlations between the Albany, Fraser and Darling mobile belts of Western Australia and Mirnyy to Windmill Islands in the East Antarctic Shield: implications for Proterozoic Gondwanaland reconstructions. *In*: Yoshida M. & Santosh M. eds. *India and Antarctica during the Precambrian*, pp. 47–71. Memoir of the Geological Society of India **34**.

HARRIS L. B. & LI Z. X. 1995. Palaeomagnetic dating and tectonic significance of dolerite intrusions in the Albany Mobile Belt, Western Australia. *Earth and Planetary Science Letters* **131**, 143–164.

IDNURM M. & GIDDINGS J. W. 1988. Australian Precambrian polar wander: a review. *Precambrian Research* **440/42**, 61–88.

IDNURM M., GIDDINGS J. W. & PLUMB K. A. 1995. Apparent polar wander and reversal stratigraphy of the Palaeo-Mesoproterozoic southeastern McArthur Basin, Australia. *Precambrian Research* **72**, 1–41.

LOWRIE W. 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. *Geophysical Research Letters* **17**, 159–162.

LOWRIE W. & FULLER M. 1971. On the alternating field demagnetisation characteristics of multidomain thermoremanent magnetization in magnetite. *Journal of Geophysical Research* **76**, 6339–6349.

MCFADDEN P. L. & MCELHINNY M. W. 1990. Classification of the reversal test in paleomagnetism. *Geophysical Journal International* **103**, 725–729.

MICHAEL S. S. 1991a. Processing of BMR magnetics, Mount Barker 1:250 000 sheet, multiclient study. Geoimage Pty Ltd. (unpubl.).

MICHAEL S. S. 1991b. Processing of BMR magnetics, Bremer Bay 1:250 000 sheet, multiclient study. Geoimage Pty Ltd. (unpubl.).

PIPER J. D. A. 1987. *Palaeomagnetism and the Continental Crust*. Open University Press, Milton Keynes.

SCHMIDT P. W. 1993. Palaeomagnetic cleaning strategies. *Physics of the Earth and Planetary Interiors* **76**, 1169–1178.

TANAKAH H. & IDNURM M. 1994. Palaeomagnetism of Proterozoic mafic intrusions and host rocks of the Mount Isa Inlier, Australia: revised. *Precambrian Research* **69**, 241–258.

TARLING D. H. & HROUDA F. 1993. *The Magnetic Anisotropy of Rocks*. Chapman & Hall, London.

THOMPSON R. & OLDFIELD F. 1986. *Environmental Magnetism*. Allen & Unwin, London.

ZIJDERVELD J. D. A. 1967. A. C. demagnetization of rocks: analysis of results. *In*: Collinson D. W., Creer K. M. & Runcorn S. K. eds. *Methods in Palaeomagnetism* , pp. 254–286. Elsevier, Amsterdam.

Received 22 October 1999; accepted 30 August 2000